CIE AS PHYSICS (9702) • QUESTION PAPER

7.4 Electromagnetic Spectrum

Syllabus Section: 7. Waves **Paper Type:** Multiple Choice (Paper 1) **Total:** 18 Questions (18 Marks)

Question 1 Easy [1 mark]

Which of the following is the approximate range of frequencies of infrared radiation?

A $1 \times 10^{14} - 1 \times 10^{16}$ Hz

B $1 \times 10^{12} - 1 \times 10^{14}$ Hz

C $1 \times 10^{9} - 1 \times 10^{11}$ Hz

D $1 \times 10^{3} - 1 \times 10^{9}$ Hz

Question 2 Easy [1 mark]

Which of the following statements is a correct description of part of the electromagnetic spectrum?

A waves of wavelength 5×10^{-7} m are infrared waves

B waves of wavelength 5×10^{-9} m are high energy gamma rays

C waves of wavelength 9×10^{-7} m are light waves

D waves of wavelength 3×10^{-8} m are ultraviolet waves

Question 3 Easy [1 mark]

Which of the following give the order of magnitude of frequencies of the visible part of the electromagnetic spectrum?

A 10^{15} Hz

B 10^{14} Hz

C 10^{13} Hz

D 10^{16} Hz

Question 4 Easy [1 mark]

The diagram shows part of the electromagnetic spectrum.



Which row correctly labels the regions marked P and Q?

	P	Q
A	infrared	x-ray
B	ultraviolet	x-ray
C	x-ray	infrared
D	infrared	microwaves

A

B

C

D

Question 5 Easy [1 mark]

Which of the following do **not** travel at the speed of light in a vacuum?

A x-rays

B radio waves

C microwaves

D electrons

Question 6 Easy [1 mark]

Which of the following is the number of wavelengths in one metre of visible light?

A 10^4

B 10^6

C 10^8

D 10^{10}

Question 7 Medium [1 mark]

Two lasers are set up to pass through a vacuum. One laser emits red light; the other emits green light.

Which property of the two lasers are different?

A speed

B plane of polarisation

C frequency

D amplitude

Question 8 Medium [1 mark]

The wavelength of light emitted by a laser is 600 nm.

Which of the following is the distance in number of wavelengths, that is travelled by light in one second?

A $5.0 \times 10^{14}\ s^{-1}$

B $5.0 \times 10^{8}\ s^{-1}$

C $5.0 \times 10^{11}\ s^{-1}$

D $5.0 \times 10^{17}\ s^{-1}$

Question 9 Medium [1 mark]

Waves of the electromagnetic spectrum travel through a vacuum at speed c and with a wavelength λ and frequency f

Which of the following describes the speed and wavelength when the frequency is $\frac{f}{2}$?

	λ	c
A	2λ	$2c$
B	2λ	c
C	$\frac{\lambda}{2}$	c
D	$\frac{\lambda}{2}$	$\frac{c}{2}$

A

B

C

D

Question 10 Medium [1 mark]

The diameter of a nucleus and the wavelength of an electromagnetic wave are in the same order of magnitude.

In which region of the electromagnetic spectrum would that wavelength occur?

A infrared

B visible light

C x-ray

D gamma-ray

Question 11 Medium [1 mark]

R is a source emitting infra-red radiation and S is a source emitting ultra-violet radiation. The table suggests the wavelengths for R and S.

Which row is correct?

	Wavelength emitted by R / m	Wavelength emitted by S / m
A	5×10^{-7}	5×10^{-8}
B	5×10^{-8}	5×10^{-10}
C	5×10^{-5}	5×10^{-8}
D	5×10^{-5}	5×10^{-10}

A

B

C

D

Question 12 Medium [1 mark]

Which statement would be deduced from a table of wavelengths of the waves in the electromagnetic spectrum?

A the wavelength range for radio waves is less than that for infrared waves

B the wavelength range for x-rays is less than that for radio waves

C red light has a shorter wavelength than green light

D green light has a shorter wavelength than x-rays

Question 13 Medium [1 mark]

A telescope detects electromagnetic radiation with wavelengths 2 cm.

Which type of telescope is it ?

A x-ray telescope

B optical telescope

C microwave telescope

D radio telescope

Question 14 Medium [1 mark]

P is a source emitting radio waves and Q is a source emitting X-rays. The table suggests the wavelengths for P and Q.

Which row is correct?

	Wavelength emitted by P / m	**Wavelength emitted by Q / m**
A	3×10^{-1}	3×10^{-11}
B	3×10^{-7}	3×10^{-15}
C	3×10^{-2}	3×10^{-15}
D	3×10^{-7}	3×10^{-11}

A

B

C

D

Question 15 Hard [1 mark]

An infrared wave has a wavelength of 30 μm and frequency f_{infrared}. An x-ray has a wavelength of 400 pm and frequency $f_{\text{x-ray}}$.

What is the value of the ratio $\frac{f_{x\text{-}ray}}{f_{infrared}}$?

A 1.3×10^{-5}

B 0.075

C 13

D 7.5×10^{4}

Question 16 Hard [1 mark]

Fig. 1.1 shows the percentage of transmitted wavelengths which can travel through Earth's atmosphere and reach sea level.

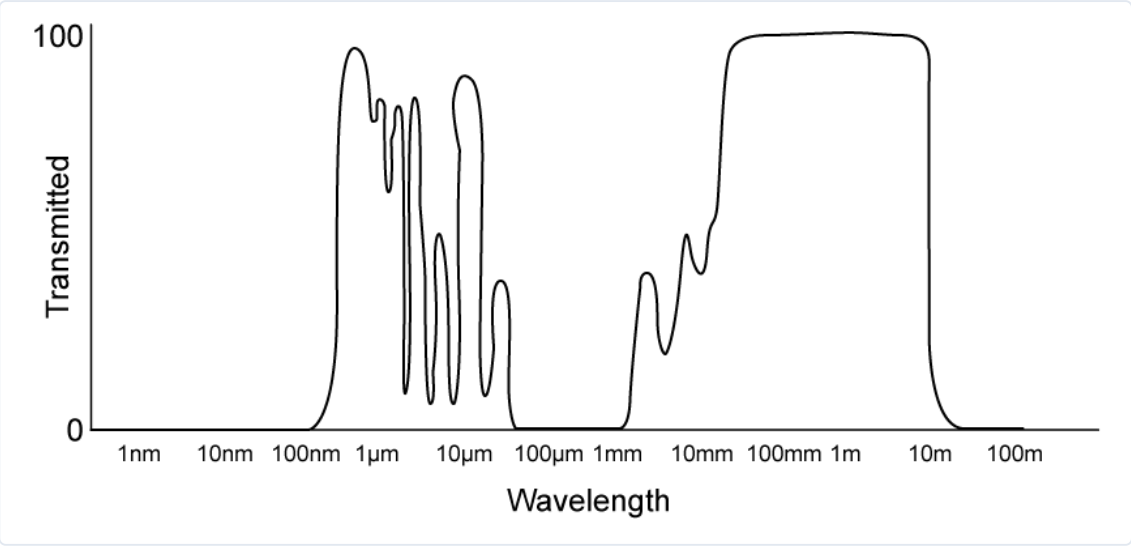


Fig. 1.1

What is the range of frequencies that can penetrate Earth's atmosphere and reach sea level?

A $10 \text{ nHz} \leq f \leq 100 \text{ μHz}, \quad 1 \text{ mHz} \leq f \leq 10 \text{ Hz}$

B $10 \text{ Hz} \leq f \leq 100 \text{ Hz}$

C $30 \text{ MHz} \leq f \leq 300 \text{ GHz}, \quad 3 \text{ THz} \leq f \leq 3000 \text{ THz}$

D $30 \text{ MHz} \leq f \leq 300 \text{ THz}$

Question 17 Hard [1 mark]

Monochromatic light is incident normally upon a double slit where the distance between the slits is 0.02 mm. The angle between the second-order and central maxima is 3.96°.

What is the colour of the monochromatic light?

A Violet

B Blue

C Green

D Red

Question 18 Hard [1 mark]

The table below contains the frequencies of various parts of the electromagnetic spectrum.

Which row correctly describes X as infrared and Y as X-rays?

	X	Y
A.	3×10^{5} Hz	3×10^{20} Hz
B.	3×10^{10} Hz	3×10^{16} Hz
C.	3×10^{18} Hz	3×10^{14} Hz
D.	3×10^{14} Hz	3×10^{19} Hz

A

B

C

D